

Analysis (Experiments in Further Pages)

The buffer size is having an impact on the runtime of the program because it affects how many syscalls are made. Each syscall requires all of the logic I wrote to preserve registers and call it, and then the processing the system does, also requiring context switching and additional potential overhead like scheduling.

50 characters is clearly too small of a size, with the only upside being that it uses only 50 bytes of space (typical space vs time trade off). However, with the current algorithm the small buffer size caused words to get chopped apart when a new buffer was requested, and not get counted as matching.

The optimal buffer size for this particular application on my machine appears to be 100,000, since the time it takes does not linearly decrease as buffer size increases. My guess as to why this is observed is that function calls with extremely large buffers take significantly longer, since both the user and sys times went up in 300,000 and 750,000 cases.

BUFFERSIZE = 50

real 0m0.058s
user 0m0.040s
sys 0m0.022s

```
Found: should - 28  
Found: should - 29  
Found: should - 30  
Found: should - 31  
Found: should - 32  
Found: should - 33  
Found: should - 34  
Found: should - 35  
Found: should - 36  
Found: should - 37  
Found: should - 38  
Found: should - 39  
Found: should - 40  
Total Instances of should Found: 40  
  
real 0m0.058s  
user 0m0.040s  
sys 0m0.022s
```

BUFFERSIZE = 100,000

```
real  0m0.024s
user  0m0.025s
sys   0m0.001s
```

```
Found: should - 33
Found: should - 34
Found: should - 35
Found: should - 36
Found: should - 37
Found: should - 38
Found: should - 39
Found: should - 40
Found: should - 41
Found: should - 42
Found: should - 43
Found: should - 44
Found: should - 45
Total Instances of should Found: 45
```

```
real  0m0.024s
user  0m0.025s
sys   0m0.001s
```

BUFFERSIZE = 300,000

```
real  0m0.033s
user  0m0.031s
sys   0m0.004s
```

```
Found: should - 32
Found: should - 33
Found: should - 34
Found: should - 35
Found: should - 36
Found: should - 37
Found: should - 38
Found: should - 39
Found: should - 40
Found: should - 41
Found: should - 42
Found: should - 43
Found: should - 44
Found: should - 45
Total Instances of should Found: 45

real    0m0.033s
user    0m0.031s
sys     0m0.004s
```

BUFFERSIZE = 750,000

```
real  0m0.035s
user  0m0.032s
sys   0m0.005s
```

```
Found: should - 33
Found: should - 34
Found: should - 35
Found: should - 36
Found: should - 37
Found: should - 38
Found: should - 39
Found: should - 40
Found: should - 41
Found: should - 42
Found: should - 43
Found: should - 44
Found: should - 45
Total Instances of should Found: 45

real    0m0.035s
user    0m0.032s
sys     0m0.005s
```